

# Mridul Sahu

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Senior Software Engineer on Google's Core ML Frameworks team, building and optimizing **Orbax**, JAX's distributed checkpointing library. I work the **JAX / XLA** training stack at the sharding, collective, and compiler level: checkpoint I/O, fault tolerance, and large-model save/load. Contributor to the **Orbax paper** (arXiv, 2026). 8+ years building high-performance distributed systems across ML infrastructure, planet-scale backend services, and low-latency trading platforms.

## WORK EXPERIENCE

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### Google

Oct. 2021 – Present

Senior Software Engineer, Core ML Frameworks (Nov. 2024 – Present)

Bangalore, India

- **Cut Orbax checkpoint blocking-save time 30–50%**, with substantial annualized TPU/GPU savings, by designing an asynchronous directory-creation mechanism on JAX distributed-client coordination; awarded Google's internal Perfy performance award.
- Designed a **sharding-driven checkpoint loader** that reads from the target `jax.sharding` rather than file layout, eliminating per-tensor reshard collectives and XLA recompilation and cutting cluster read amplification to  $\sim 1\times$  the checkpoint.
- Cut large-model restart time and **storage read amplification  $N\times \rightarrow 1\times$**  with Single-Replica Restore + Broadcast (one replica loads, then broadcasts); hardened fault tolerance via the Emergency Checkpoint Manager's two-stage local + persistent save.
- Architected Orbax's next-generation **Dispatcher API**, unifying multi-controller and single-controller checkpointing behind one extensible interface that lets single-controller frameworks like Pathways checkpoint at large distributed scale.
- Built the **Knowledge Distillation** module for Tunix (Google's JAX-native LLM post-training library): a strategy-pattern framework for logit, feature/attention transfer, and pooling/projection alignment across heterogeneous architectures.
- Eliminated Tunix per-step **XLA recompilation** by hoisting `nnx.jit` to one-time setup, and fixed TPU HBM OOM on load via CPU host placement.
- Built Orbax's checkpoint **benchmarking and observability framework** (stage timings, XProf traces, per-device HBM usage, TensorStore I/O) with A/B regression baselines, correctness digests, and CI/CD; validated at llama3-8B/70B/405B scale.
- Resolved high-stakes Orbax performance and stability escalations for big-tech Google Cloud customers, unblocking training runs of models with **billions to trillions of parameters** on fleets of 32k+ TPU chips.
- Founding member of the **Core ML India** team; mentor 8+ engineers across the Keras, JAX, and Orbax ecosystems.

Software Engineer III, Google One (Oct. 2021 – Oct. 2024)

Bangalore, India

Storage Quota Service · Apr. 2023 – Oct. 2024

- Designed an asynchronous processing system that eliminated database lock contention and sustained **O(100K) write QPS**, unlocking usage-calculation capacity across all Google services.
- Tech-led a Gmail  $\times$  Storage Quota Service initiative for a **billion-plus-user** base: built a scalable experimentation framework, piloted a 30-day storage grace period in Gmail's codebase, and shipped email + in-Gmail warning banners alerting millions of users.
- Shipped a phased out-of-storage launch reaching millions of users each year with **zero production incidents**, driving substantial Google One signups and upgrades.
- Built a data-validation and recovery pipeline safeguarding storage-usage integrity for **billions of user accounts**, plus a precision outage-recovery mechanism restoring usage across Gmail, Photos, and Drive.

- Built real-time **service-level usage-breakdown** infrastructure for Enterprise Customers, eliminating inefficient caching of millions of user-level rows in admin consoles.

*Benefits · Oct. 2021 – Apr. 2023*

- Built the foundational benefits-eligibility backend (cornerstone of a major customer-acquisition feature), improving performance **70% over v1**, plus a reconciliation framework preventing data inconsistencies across partner teams (Photos, YouTube, Workspace, GStore).
- Led multiple benefit launches and international expansions impacting **millions of users** as central technical liaison across Product Areas; built backend to dynamically split/combine benefits for tailored delivery.
- Fixed critical infrastructure deficiencies affecting millions of users; established alerting/SLO/monitoring standards; owned a **zero-incident on-call migration** of the Benefits and Storage Quota Service teams from the US to India.

## Zopte Technologies (CrispyGames)

**2018 – 2021**

*Software Engineer*

*Mumbai, India*

- Engineered a Python backtesting and live-trading platform (Bloomberg, Zerodha, NSE-NNF, QuickFIX, custom TimeSeries DB) with an **~80× performance uplift**, plus a distributed backtesting optimizer for strategy-parameter search over a compute cluster.
- Built a low-level **NSE/NFO exchange** library (proprietary protocols, binary structures, security compliance), an Order Management System (OMS), and Tick Collector; maintained an NSE-licensed CTCL (computer-to-computer link) trading application in production at a brokerage.
- Built a code generator producing customizable, mockable server- and client-side codebases (up to **60% of project code**), wired to a custom real-time continuous-deployment system.
- Architected a **microservices service mesh** from scratch on Kubernetes with GCP and Istio (gRPC/REST, Zookeeper, Redis, Elasticsearch, MongoDB); built pluggable Flutter / Angular Dart UI components.

## PROJECTS

[research.rudrite.com](https://research.rudrite.com) · Interactive Research-Paper Visualizer

**2025 – Present**

- Building a site that **visualizes and explains landmark AI/ML research papers** through 120+ interactive, animated explainers spanning training-at-scale parallelism (Megatron, GSPMD, ZeRO), efficient LLM serving (batching, speculation, KV-cache), transformers, and RLHF / reasoning.

[explainers.rudrite.com](https://explainers.rudrite.com) · Interactive Visual Explainers

**2026 – Present**

- Building a library of **30+ interactive visual explainers** of system design, distributed systems, and real code, from high-level architecture to line-by-line code walks.

## PUBLICATIONS

- C. Gaffney, S. Li, D. Ng, **M. Sahu**, et al., "**Orbax: Distributed Checkpointing with JAX**," *arXiv:2605.23066* (cs.DC / cs.LG), 2026. A modular, JAX-native checkpointing library demonstrating up to **3.5x faster saving and 2x faster loading** vs. comparable PyTorch checkpointing.
- S. K. Jaiswal, M. Pal, **M. Sahu**, P. Sahu, A. Dev, "EvoCut: A New Generalization of the Albert-Barabási Model for Evolution of Complex Networks," *22nd FRUCT Conference*, Jyväskylä, 2018, pp. 67–72.

## EDUCATION

**National Institute of Technology, Meghalaya**

**2014 – 2018**

*B.Tech, Computer Science & Engineering, CGPA 8.76/10*

*Shillong, India*

## TECHNICAL SKILLS

**Languages:** Python, Java, Go, C++, Dart, JavaScript

**ML & Performance:** JAX, XLA, Orbax, Flax (NNX), Optax, Tunix, TensorStore, Pathways on Cloud; distributed training, sharding, collectives, TPU, profiling (XProf, jax.profiler, jax.monitoring), knowledge distillation, ML systems

**Systems & Infrastructure:** distributed systems, microservices, system design, API design, low-latency / trading systems (FIX, OMS, exchange protocols), gRPC, REST; Kubernetes, GCP, Docker, XPK

**Data & Web:** SQL, MySQL, MongoDB, Elasticsearch, Redis, Cloud Pub/Sub; Flutter, Angular Dart, OpenCV